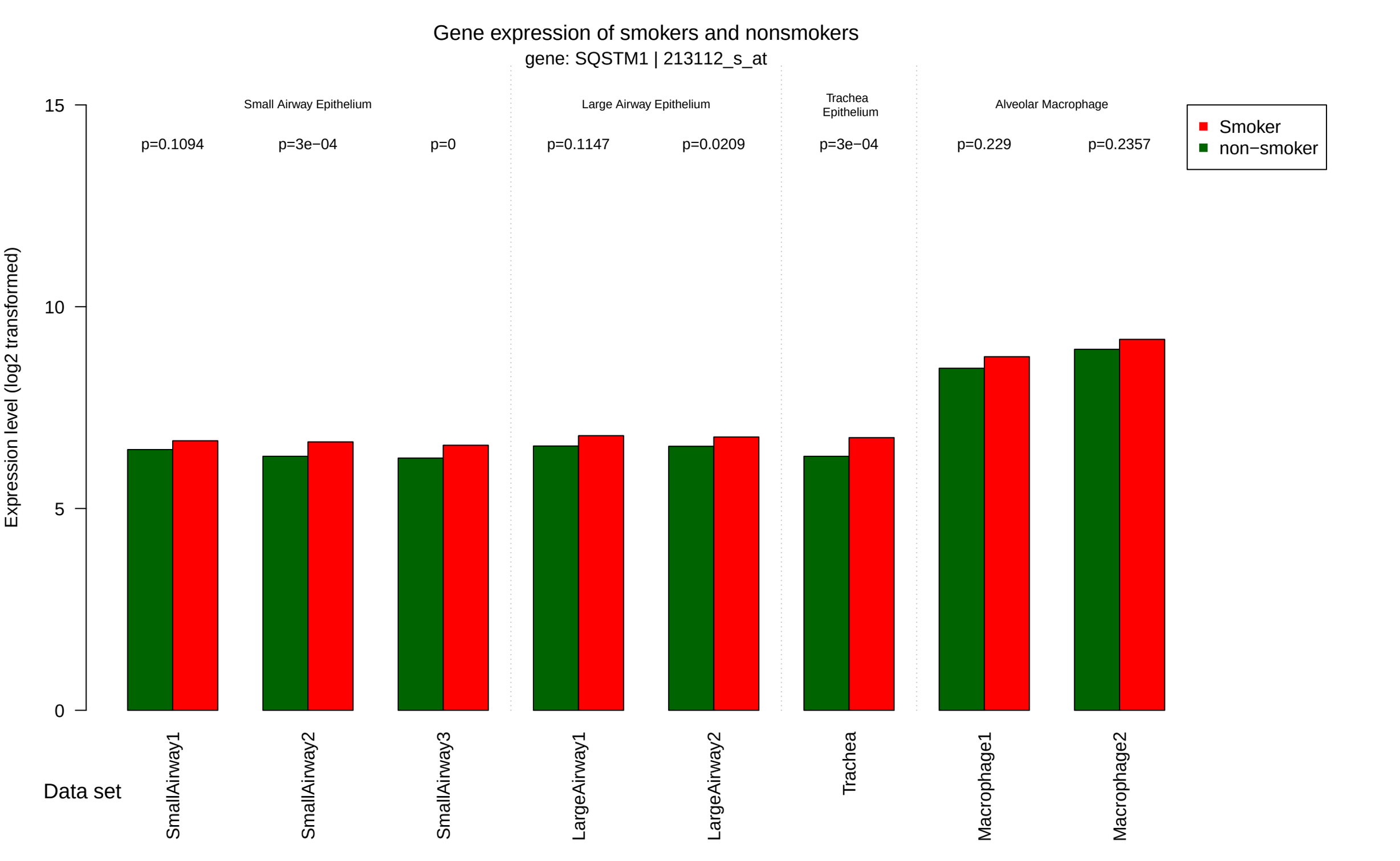
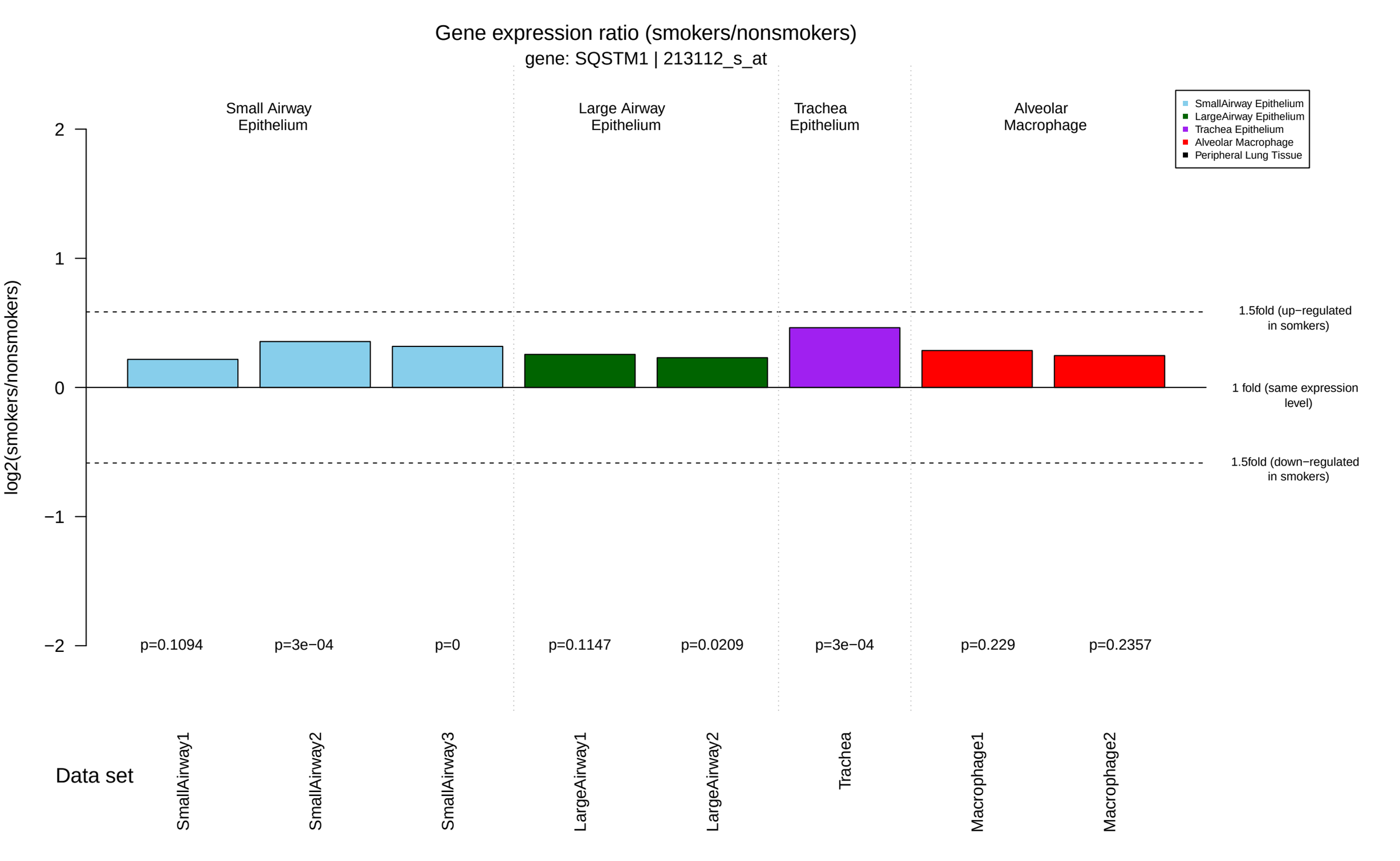
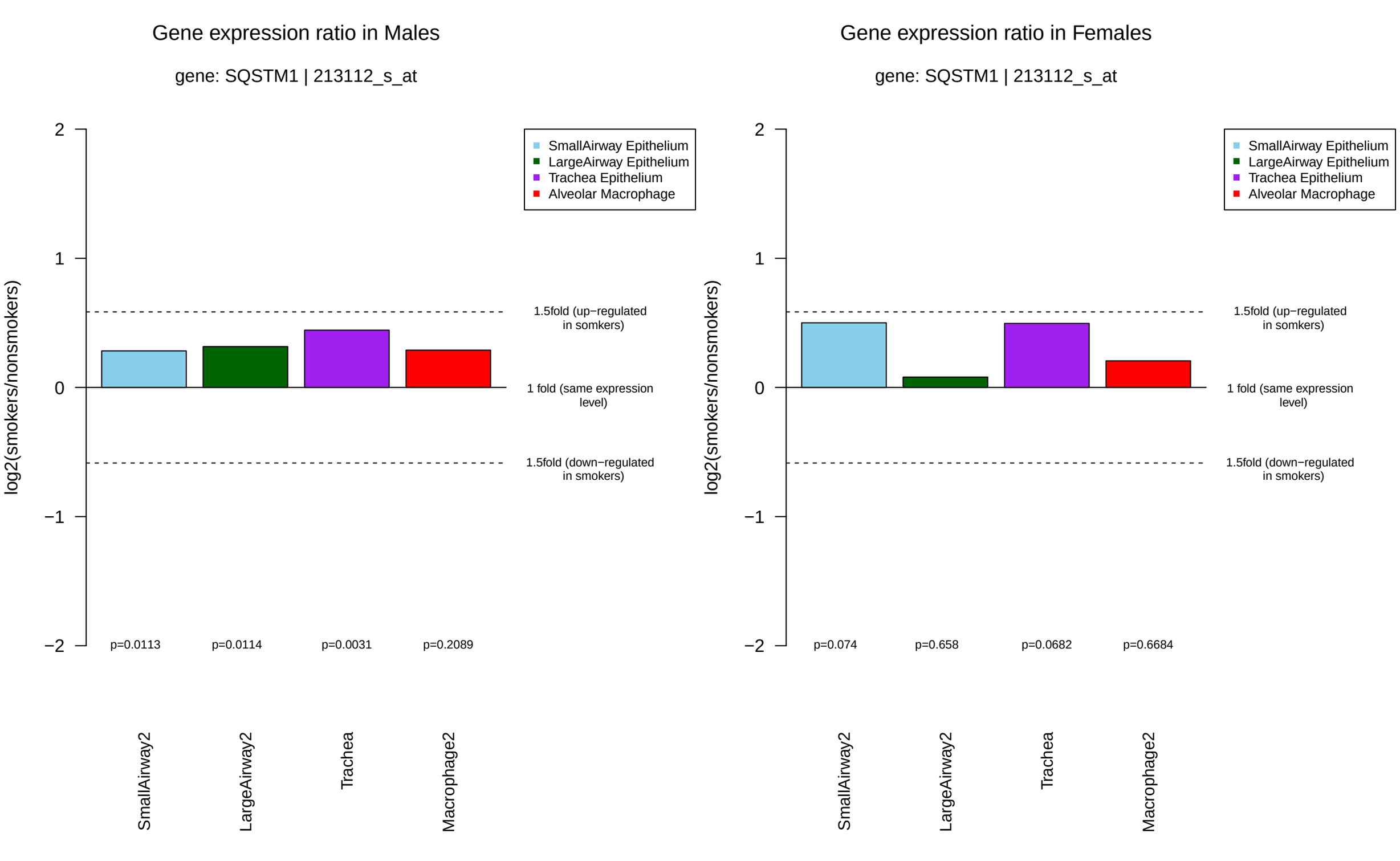
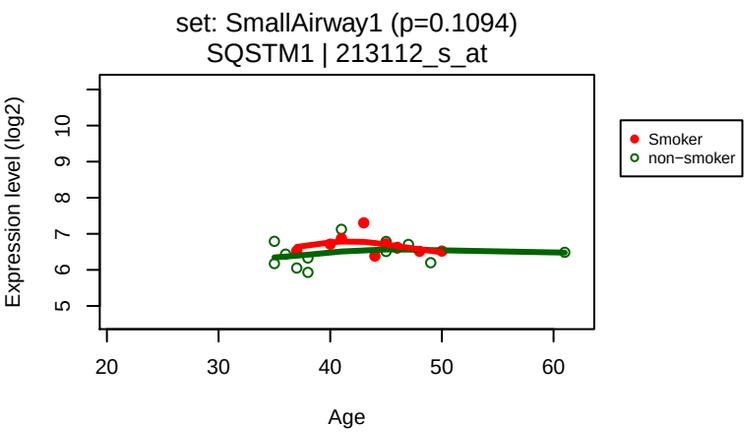
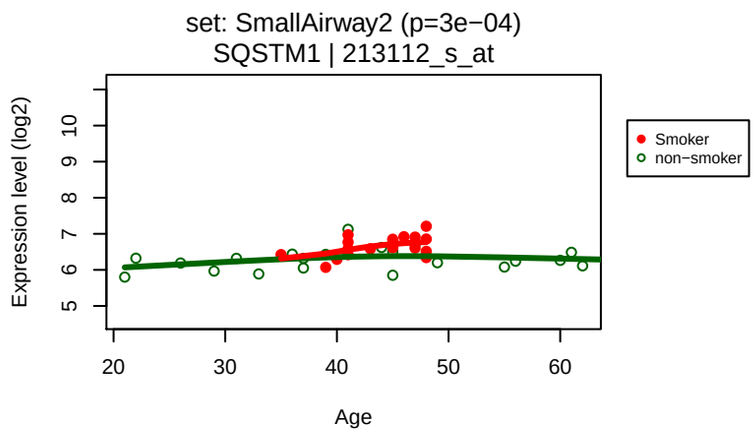
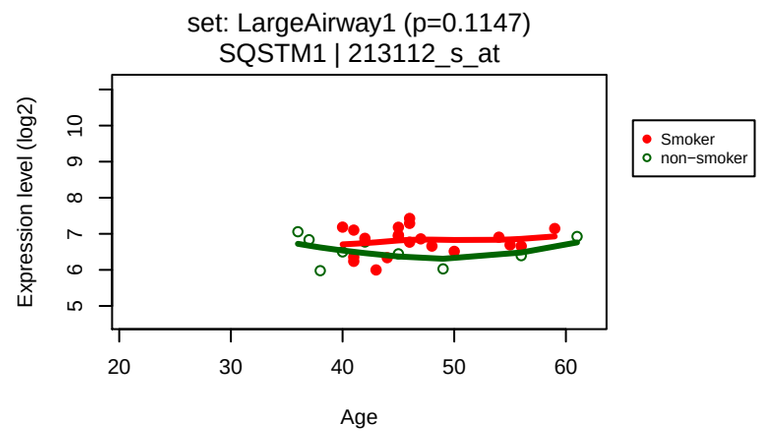

Gene expression of smokers and nonsmokers
gene: SQSTM1 | 213112_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1094
p=3e−04
p=0
p=0.1147
p=0.0209
p=3e−04
p=0.229
p=0.2357
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

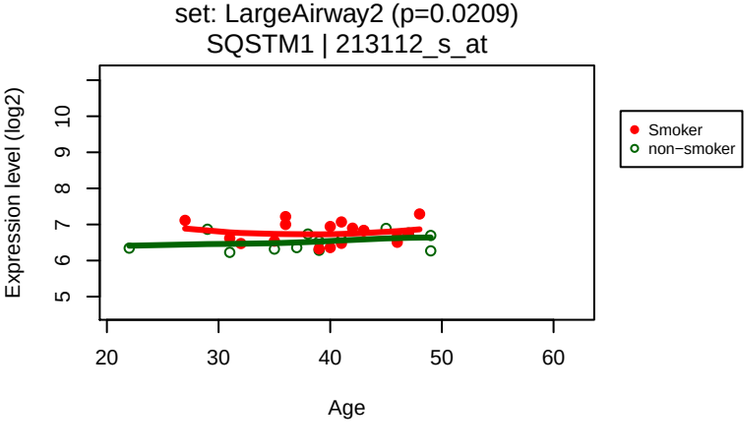

Gene expression ratio (smokers/nonsmokers)
gene: SQSTM1 | 213112_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1094
p=3e−04
p=0
p=0.1147
p=0.0209
p=3e−04
p=0.229
p=0.2357
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

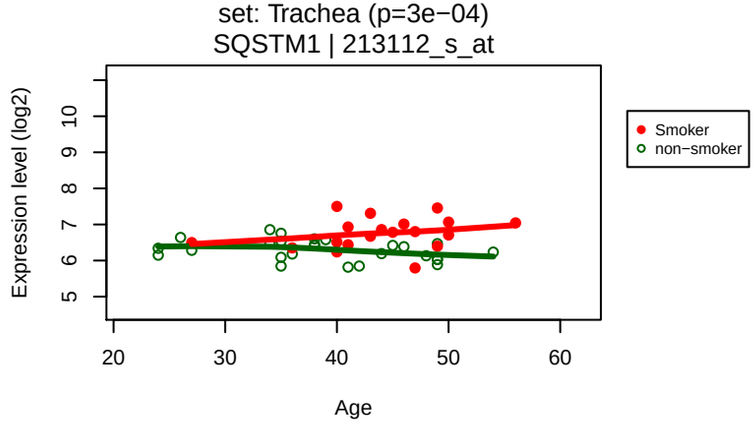

Gene expression ratio in Males
gene: SQSTM1 | 213112_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0113
p=0.0114
p=0.0031
p=0.2089
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SQSTM1 | 213112_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.074
p=0.658
p=0.0682
p=0.6684
SmallAirway2
LargeAirway2
Trachea
Macrophage2

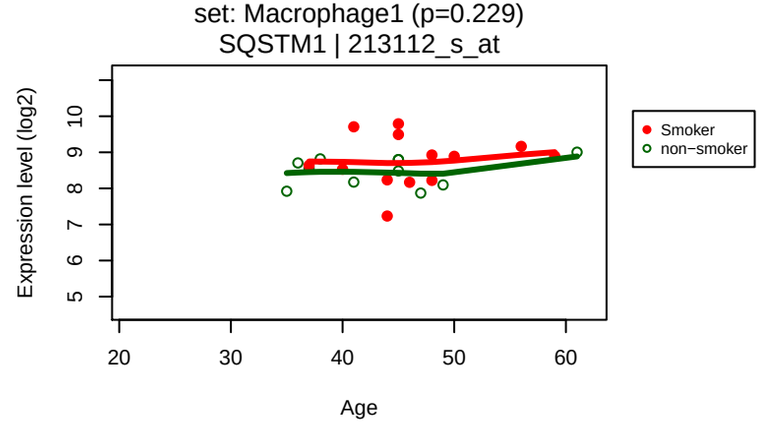

set: SmallAirway1 (p=0.1094)
SQSTM1 | 213112_s_at
Expression level (log2)
Age
Smoker
non−smoker

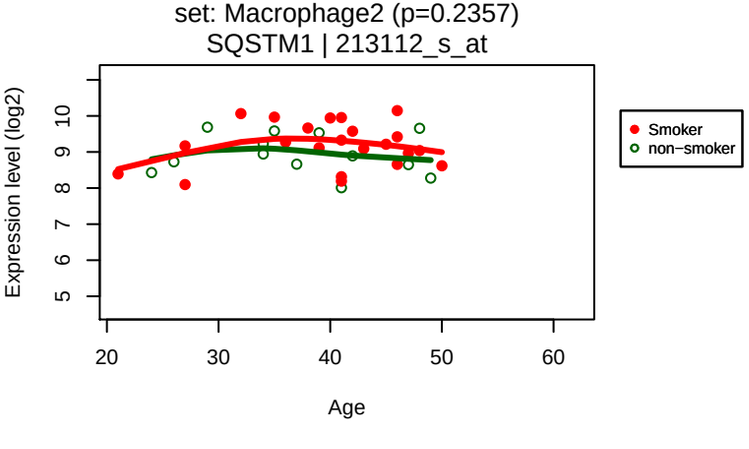

set: SmallAirway2 (p=3e−04)
SQSTM1 | 213112_s_at
Expression level (log2)
Age
Smoker
non−smoker

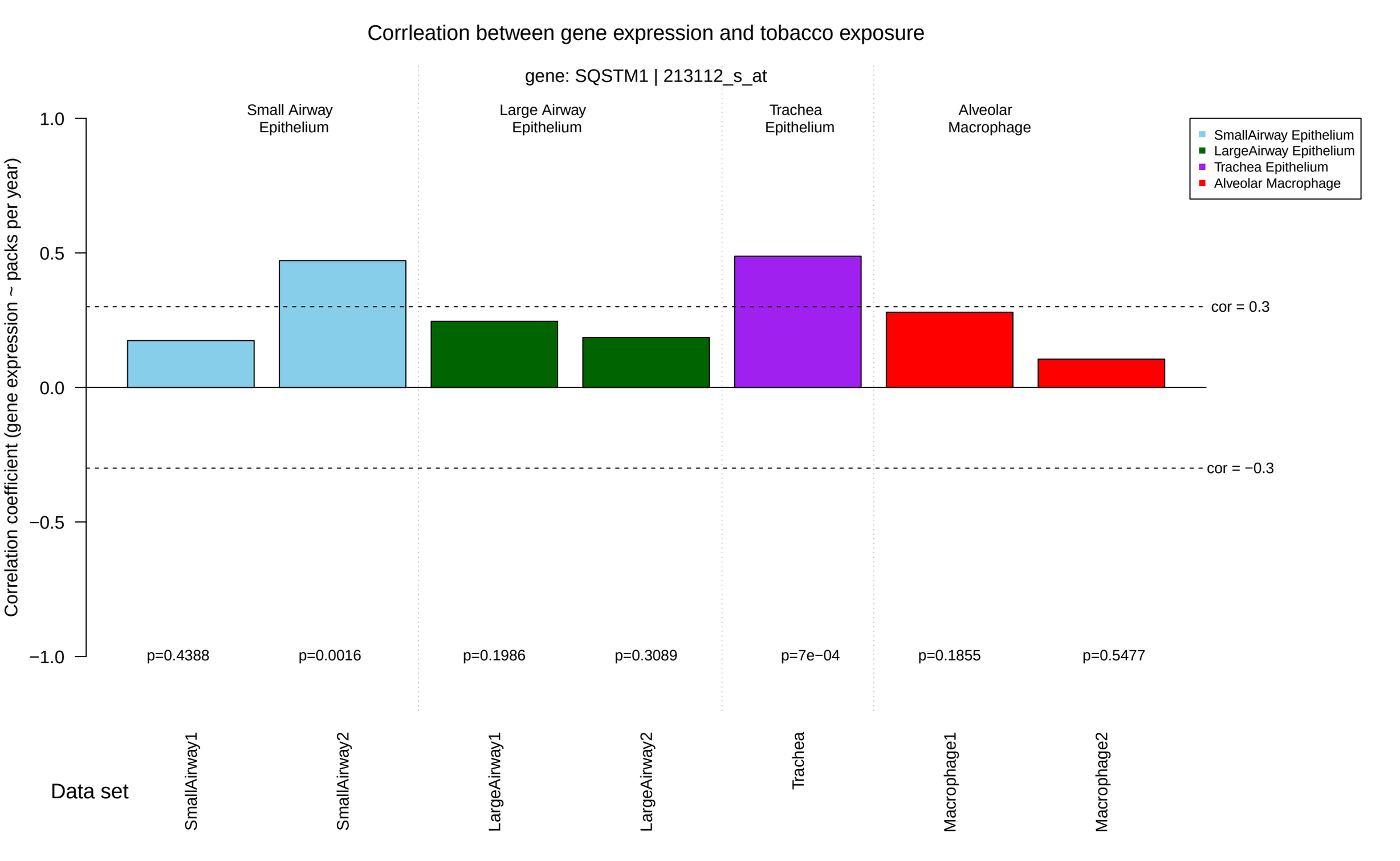

set: LargeAirway1 (p=0.1147)
SQSTM1 | 213112_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0209)
SQSTM1 | 213112_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=3e−04)
SQSTM1 | 213112_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.229)
SQSTM1 | 213112_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2357)
SQSTM1 | 213112_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SQSTM1 | 213112_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4388
p=0.0016
p=0.1986
p=0.3089
p=7e−04
p=0.1855
p=0.5477
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2